

Week 12 - Global Energy: Worked Answers

EART23001 Atmospheric Physics and Weather

Workload guide. The shaded question headings show the intended workload:

- **IN CLASS:** selected anchor problems that we may work through together in the timetabled session.
- **YOU SHOULD TRY:** expected independent practice after the session.
- **EXTRA CHALLENGE:** optional problems for students who want additional depth or a harder application.

The categories are repeated here so the worked answers map directly onto the practice sheet. Use the solutions to check method, units and physical interpretation rather than only the final number. A course-wide Symbols, Constants and Units sheet is provided separately; non-standard symbols are also defined locally.

Useful constants: solar luminosity $L_{\odot} = 3.828 \times 10^{26}$ W, mean Earth–Sun distance $d = 1.496 \times 10^{11}$ m, Stefan–Boltzmann constant $\sigma_{\text{SB}} = 5.67 \times 10^{-8}$ W m⁻² K⁻⁴, Wien constant $b = 2.898 \times 10^{-3}$ m K.

YOU SHOULD TRY Question 1: From solar luminosity to the solar constant

Assume the Sun radiates isotropically.

1. Calculate the solar irradiance at the mean Earth–Sun distance using

$$S_0 = \frac{L_{\odot}}{4\pi d^2}.$$

2. Why is the global-mean incoming solar flux at the top of the atmosphere $S_0/4$ rather than S_0 ?
3. Calculate $S_0/4$.

Worked answer

$$S_0 = \frac{3.828 \times 10^{26}}{4\pi(1.496 \times 10^{11})^2} = 1.361 \times 10^3 \text{ W m}^{-2}.$$

The Earth intercepts sunlight over a disk of area πR^2 , but the intercepted energy is averaged over the full spherical area $4\pi R^2$. Therefore

$$\langle F_{\text{SW}} \rangle = \frac{S_0 \pi R^2}{4\pi R^2} = \frac{S_0}{4} \approx 340 \text{ W m}^{-2}.$$

IN CLASS Question 2: Earth's effective radiating temperature

Take Earth's planetary albedo to be $\alpha = 0.30$. In a simple zero-dimensional radiative equilibrium,

$$\frac{(1 - \alpha)S_0}{4} = \sigma_{\text{SB}} T_e^4.$$

1. Calculate the globally averaged absorbed solar flux.
2. Calculate T_e .

3. Compare it with a representative global-mean surface temperature of 288 K and calculate the difference.
4. As a simple grey diagnostic, write the outgoing longwave flux as

$$F_{out} = \epsilon_{eff} \sigma_{SB} T_s^4.$$

Using $T_s = 288$ K, calculate the effective emissivity ϵ_{eff} required for $F_{out} = F_{abs}$. Then substitute your value back into

$$T_s = \left(\frac{F_{abs}}{\epsilon_{eff} \sigma_{SB}} \right)^{1/4}$$

to verify that $\epsilon_{eff} \simeq 0.61$ corresponds to $T_s \simeq 288$ K. Is this the emissivity of the physical surface itself?

Worked answer

The absorbed solar flux is

$$F_{abs} = \frac{0.70(1361)}{4} = 238.2 \text{ W m}^{-2}.$$

Hence

$$T_e = \left(\frac{238.2}{5.67 \times 10^{-8}} \right)^{1/4} = 254.6 \text{ K}.$$

The representative surface temperature exceeds this by

$$288 - 254.6 = 33.4 \text{ K}.$$

If we represent the whole Earth–atmosphere system by an effective longwave emissivity referenced to the surface temperature, then

$$\epsilon_{eff} = \frac{F_{out}}{\sigma_{SB} T_s^4} = \frac{238.2}{(5.67 \times 10^{-8})(288)^4} \approx 0.61.$$

Substituting this value back into the grey diagnostic gives

$$T_s = \left[\frac{238.2}{0.61(5.67 \times 10^{-8})} \right]^{1/4} \approx 288 \text{ K}.$$

So, in this deliberately simple representation, reducing the effective longwave emissivity from unity to about 0.61 raises the temperature required to emit 238.2 W m^{-2} from 254.6 K to about 288 K. This is *not* the emissivity of Earth's physical surface, which is close to unity over much of the thermal infrared. It is an effective property of the surface–atmosphere system, reflecting that much of the radiation reaching space is emitted from colder atmospheric levels.

IN CLASS Question 3: Cloud droplet number, M_2 , extinction and albedo

Cloud A is $\Delta z = 500$ m deep and contains $N_A = 100 \text{ cm}^{-3}$ droplets of radius $r_A = 10 \text{ }\mu\text{m}$. Cloud B has the same liquid-water content and depth, but $N_B = 800 \text{ cm}^{-3}$. Assume identical droplets within each cloud.

For this simplified exercise define the area-related second moment as

$$M_2 = N r^2,$$

and use the approximate cloud-albedo relation

$$A_c \approx \frac{X}{X + 15.4}, \quad X = \pi M_2 \Delta z.$$

Cloud-droplet extinction is approximately proportional to an area measure such as M_2 .

1. Use constant liquid-water content ($Nr^3 = \text{constant}$) to calculate r_B .
2. Convert the concentrations to m^{-3} and calculate M_2 for both clouds.
3. By what factor does M_2 change? What does this imply for extinction?
4. Calculate X and A_c for each cloud.
5. Why does doubling M_2 not double cloud albedo?
6. Show that, at fixed liquid-water content, $M_2 \propto N^{1/3}$. Hence estimate the fractional change in M_2 for a small 30% increase in N .
7. State the likely sign of the immediate shortwave effect on Earth's energy balance if a cloud becomes brighter in this way, all else equal.

Worked answer

Fixed liquid-water content implies

$$N_A r_A^3 = N_B r_B^3,$$

so

$$r_B = 10 \left(\frac{100}{800} \right)^{1/3} = 5.0 \text{ } \mu\text{m}.$$

The concentrations are $N_A = 1.00 \times 10^8 \text{ m}^{-3}$ and $N_B = 8.00 \times 10^8 \text{ m}^{-3}$. Therefore

$$M_{2,A} = N_A r_A^2 = (1.00 \times 10^8)(10^{-5})^2 = 1.00 \times 10^{-2} \text{ m}^{-1},$$

and

$$M_{2,B} = (8.00 \times 10^8)(5.0 \times 10^{-6})^2 = 2.00 \times 10^{-2} \text{ m}^{-1}.$$

Thus M_2 doubles, so the extinction coefficient also increases by approximately a factor of two in this simple comparison.

For Cloud A,

$$X_A = \pi(0.0100)(500) = 15.71, \quad A_{c,A} = \frac{15.71}{15.71 + 15.4} = 0.505.$$

For Cloud B,

$$X_B = \pi(0.0200)(500) = 31.42, \quad A_{c,B} = \frac{31.42}{31.42 + 15.4} = 0.671.$$

So changing how the same liquid-water mass is divided among droplets raises the albedo from about 0.51 to 0.67 in this simplified example. Albedo does not double because the relation saturates: once a cloud is already optically thick/bright, additional extinction produces progressively smaller increases in reflected fraction.

At fixed liquid-water content,

$$Nr^3 = \text{constant} \quad \Rightarrow \quad r \propto N^{-1/3}.$$

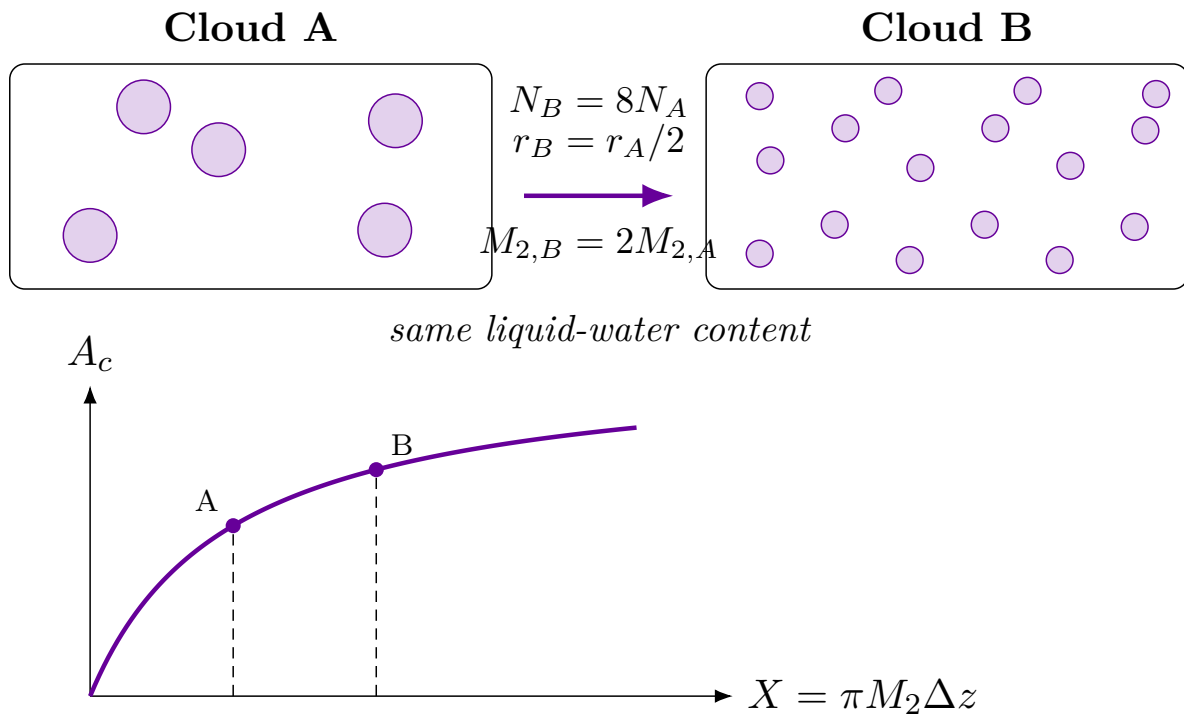
Hence

$$M_2 = Nr^2 \propto N N^{-2/3} = N^{1/3}.$$

For a small fractional perturbation,

$$\frac{\Delta M_2}{M_2} \approx \frac{1}{3} \frac{\Delta N}{N}.$$

A 30% increase in N therefore gives approximately a 10% increase in M_2 (the exact factor is $1.3^{1/3} \approx 1.091$, or about 9%). A brighter cloud reflects more incoming shortwave radiation to space, producing an immediate cooling influence on the planetary energy balance, all else equal.



The figure summarises the chain $M_2 \uparrow \Rightarrow \text{extinction} \uparrow \Rightarrow A_c \uparrow$. The albedo curve flattens as optical thickness increases, so an already bright cloud has lower albedo susceptibility.

Connection to size distributions. Real clouds contain a spectrum of droplet sizes. More formal treatments define moments by integrating over that distribution. The identical-droplet calculation above is a deliberately simple way to expose the same physics: number is associated with a zeroth-moment idea, area/extinction with a second-moment idea, and liquid-water mass with a third-moment idea.

YOU SHOULD TRY Question 4: Separate solar shortwave from terrestrial longwave

Use Wien's law to calculate the peak-emission wavelengths of

1. the Sun at 5778 K;
2. the Earth at 288 K.

Then identify the broad spectral region of each peak and state why this separation is important for the atmosphere.

Worked answer

For the Sun,

$$\lambda_{max} = \frac{2.898 \times 10^{-3}}{5778} = 5.02 \times 10^{-7} \text{ m} = 0.50 \text{ } \mu\text{m},$$

in the visible part of the solar/shortwave spectrum.

For Earth,

$$\lambda_{max} = \frac{2.898 \times 10^{-3}}{288} = 1.01 \times 10^{-5} \text{ m} = 10.1 \text{ } \mu\text{m},$$

in the thermal infrared/longwave spectrum.

Atmospheric absorption depends strongly on wavelength. The atmosphere can therefore transmit much incoming visible radiation while absorbing and re-emitting substantial terrestrial infrared radiation.

IN CLASS Question 5: A one-layer greenhouse model

Consider a deliberately simple model atmosphere that is transparent to solar radiation but absorbs *all* terrestrial longwave radiation. The atmosphere emits as a blackbody equally upward and downward. Use the absorbed solar flux from Question 2, $F_{abs} = 238.2 \text{ W m}^{-2}$.

1. At the top of the atmosphere, radiative equilibrium requires $\sigma_{\text{SB}} T_a^4 = F_{abs}$. Calculate the atmospheric emitting temperature T_a .
2. The surface receives absorbed sunlight plus downward atmospheric longwave radiation. Show that its balance is

$$\sigma_{\text{SB}} T_s^4 = F_{abs} + \sigma_{\text{SB}} T_a^4 = 2F_{abs}.$$

Calculate T_s .

3. Compare this T_s with the effective radiating temperature from Question 2 and with the real mean surface temperature of about 288 K.
4. What does the fact that the one-layer model is too warm tell you about its assumptions?

Worked answer

At the top of the atmosphere,

$$T_a = \left(\frac{238.2}{\sigma_{\text{SB}}} \right)^{1/4} = 254.6 \text{ K},$$

the same effective radiating temperature found previously.

At the surface,

$$T_s = \left(\frac{2(238.2)}{\sigma_{\text{SB}}} \right)^{1/4} = 302.8 \text{ K}.$$

Equivalently, $T_s = 2^{1/4} T_e$.

The simple absorbing atmosphere therefore raises the surface temperature above T_e , demonstrating the greenhouse mechanism through absorption and downward re-emission. However, 303 K is warmer than the real global mean of about 288 K. Real atmospheres are not single blackbody layers: longwave absorption is wavelength-dependent, some radiation escapes through the atmospheric window, clouds and vertical temperature structure matter, and non-radiative sensible/latent heat fluxes also redistribute energy.

EXTRA CHALLENGE Question 6: Venus: high albedo but a very hot surface

Venus is $1.08 \times 10^{11} \text{ m}$ from the Sun and has planetary albedo 0.70. For this question use solar luminosity $3.85 \times 10^{26} \text{ W}$.

1. Calculate the solar irradiance at Venus.
2. Calculate its simple effective radiating temperature.
3. The actual Venusian surface temperature is about 730 K. Calculate how much hotter this is than the effective radiating temperature.
4. Explain why the surface can be so hot despite the high albedo.

Worked answer

$$S_V = \frac{3.85 \times 10^{26}}{4\pi(1.08 \times 10^{11})^2} = 2.63 \times 10^3 \text{ W m}^{-2}.$$

Then

$$T_{e,V} = \left[\frac{(1 - 0.70)S_V}{4\sigma_{\text{SB}}} \right]^{1/4} = 243 \text{ K}.$$

The surface is therefore roughly

$$730 - 243 = 487 \text{ K}$$

warmer than the simple effective radiating temperature. Venus has a massive CO₂ atmosphere with very large infrared optical depth. Radiation escaping to space originates from high, relatively cold atmospheric levels, while the deep atmosphere and surface become extremely hot. High cloud albedo reduces absorbed sunlight but does not eliminate this strong longwave greenhouse effect.

EXTRA CHALLENGE Question 7: Use radiative imbalance to infer heat transport

At one latitude band, the annual-mean absorbed solar flux is 285 W m^{-2} and outgoing longwave radiation is 250 W m^{-2} . At a higher-latitude band, the corresponding values are 170 and 210 W m^{-2} .

1. Calculate the net top-of-atmosphere radiative imbalance in each band, taking positive as net energy gain.
2. Which band has an energy surplus and which a deficit?
3. In which direction must the climate system transport energy on average to prevent indefinite warming/cooling of the two bands?
4. Name two atmospheric/oceanic mechanisms that contribute to this transport.

Worked answer

For the lower-latitude band,

$$285 - 250 = +35 \text{ W m}^{-2},$$

so there is a radiative surplus. For the higher-latitude band,

$$170 - 210 = -40 \text{ W m}^{-2},$$

so there is a deficit.

Energy must therefore be transported poleward, from the surplus region toward the deficit region. Atmospheric mean circulations and transient mid-latitude weather systems transport sensible/latent energy, while ocean currents provide another major poleward heat transport.

YOU SHOULD TRY Question 8: Classify climate feedbacks from their causal chains

For each causal chain, classify the response as primarily *positive feedback*, *negative feedback*, or *sign depends on details*. Give a one-sentence explanation.

1. Warming → less snow/sea ice → lower albedo → more absorbed sunlight.
2. Warming → greater atmospheric water-vapour amount at similar RH → stronger infrared absorption.
3. Warming → larger thermal infrared emission to space.
4. Warming → a change in cloud amount, height, optical depth and phase.

Worked answer

1. Positive: reduced ice/snow lowers albedo and increases absorbed solar energy, amplifying the initial warming.
2. Positive: increased water vapour strengthens the greenhouse effect and amplifies warming.
3. Negative: the Planck response means a warmer planet emits more longwave radiation to space, opposing the initial imbalance.
4. Sign depends on details: clouds both reflect shortwave radiation and absorb/emit longwave radiation, so the net feedback depends on how cloud properties change.

EXTRA CHALLENGE Question 9: Match the Milankovitch cycle to its timescale and effect

Match each orbital variation to the most appropriate timescale from the list {26 000, 41 000, 100 000} years and state what aspect of Earth's orbit/orientation changes.

1. eccentricity;
2. obliquity (axial tilt);
3. precession.

Why can these cycles alter regional/seasonal climate even though they do not simply act like a permanent increase in the Sun's luminosity?

Worked answer

Eccentricity varies on roughly 100 000-year timescales and changes the shape of Earth's orbit. Obliquity varies on roughly 41 000 years and changes the axial tilt. Precession operates on roughly 26 000 years and changes the orientation of Earth's rotation axis/orbit relative to the seasons.

These cycles redistribute incoming solar radiation by season and latitude rather than acting as a simple uniform increase in solar output. Their climate influence therefore comes from changing seasonal and geographical contrasts and the feedbacks those changes trigger.